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Introduction to ML

Rules

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1. 75% attendance mandatory as per university rules
2. No late entries permitted
3. Laptop screens down is the default state unless specified otherwise
4. Using phones, smartwatches, earphones or any other electronic devices is NOT permitted in class unless instructed otherwise.
5. Students must occupy the front seats first, class would not start if the front seats are vacant
6. When you have a doubt in class, raise your hand, stand up when pointed by the instructor and then ask!
7. Be respectful to everyone.

* This is not an exhaustive list of rules…

Join the Lecture on NS Portal :)

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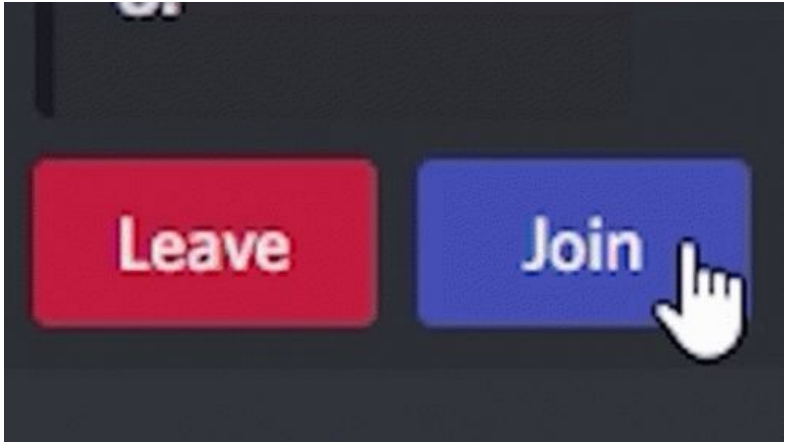

Let’s Talk

- uses AI in coding interviews
  - https://www.reddit.com/r/leetcode/comments/1o47lk2/officially_live_metas_new_aienabled_coding_round/
  - https://www.404media.co/meta-is-going-to-let-job-candidates-use-ai-during-coding-tests/

Agenda

- Introduction to Machine Learning
- Applications
- Types of Machine Learning Algorithms
- Types of Machine Learning Tasks

4-Step Learning Path

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1. **Foundation: What is ML**

   Understanding the core principles of machine learning and how it fundamentally differs from traditional programming approaches.

2. **AI → ML → DL Hierarchy**

   Exploring the relationship between artificial intelligence, machine learning, and deep learning, and how they interconnect.

4-Step Learning Path

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3. Types of Learning Methods

Discovering supervised, unsupervised, semi-supervised, and reinforcement learning approaches and their applications.

4. Core ML Tasks

Mastering classification, regression, and clustering tasks with real-world examples and practical exercises.

Is this email Spam?

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SDE-2 (MERN + Python) Role — Just Went Live Spam ×

Stuti from Abekus stuti@abekus.co.in via amazonses.com 4:20 AM (13 hours ago)
to me

Hey there,

A **Software Development Engineer 2 (MERN + Python)** role with **Mple.ai** just went **live today** on Abekus. This is a **full-time onsite** role with a compensation range of **₹10L–₹15L per year**.

Important: This job is part of an active **Hiring Event**, where companies begin reviewing candidates early.

If you have experience with the **MERN stack, Python, cloud platforms (AWS/GCP/Azure), and Docker**, **applying on the first day significantly improves your chances of getting shortlisted**, as screening starts immediately.

👉 **View & Apply Job Role Here:**
Software Development Engineer 2 (MERN + Python) – Mple.ai

Cheers,
Stuti
Team Abekus

P.S. Mark this email as important to ensure you never miss an important job update again.

Reply Forward

Finding Pattern

- Can you use if-else to solve this problem?
- A “simple” program might not work well.
- We need a system to identify patterns in suspicious words, links, and sender behaviour
- Automatically filters future emails based on learned patterns

Step 1: Introduction to Machine Learning

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Machine Learning (ML)

Computers learn patterns from data instead of being explicitly programmed with rules.

Simple Example

Imagine predicting house prices. Instead of manually writing "If 2 bedrooms → price = X", the ML model learns patterns from existing house price data and discovers the relationships automatically.

Pattern Recognition Vs Hard Coding

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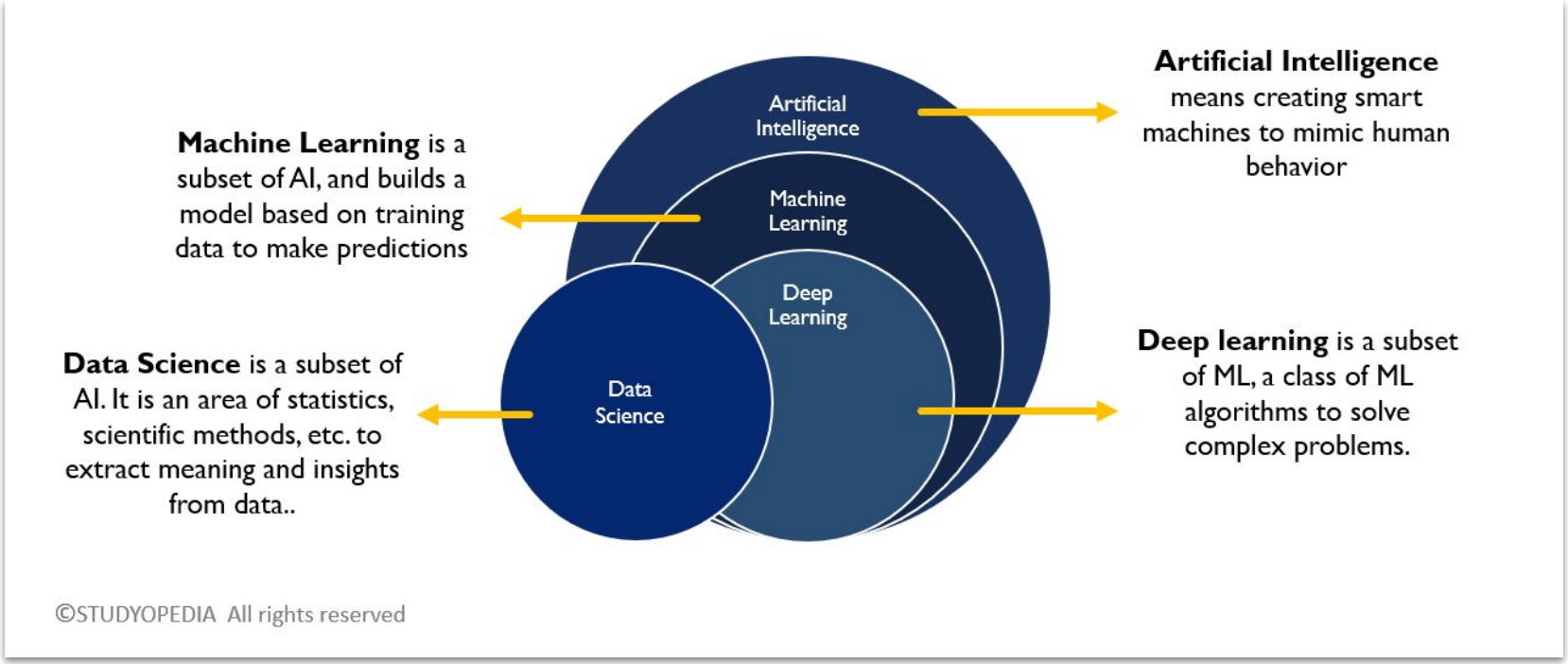
- Let’s write the number **4**.

Types of algorithms

- Deterministic
- Non Deterministic
- Probabilistic

Step 2: Understanding AI, ML, and DL



Source: https://studyopedia.com/data-science/difference-datascience-machinelearing-ai-dl/#google_vignette

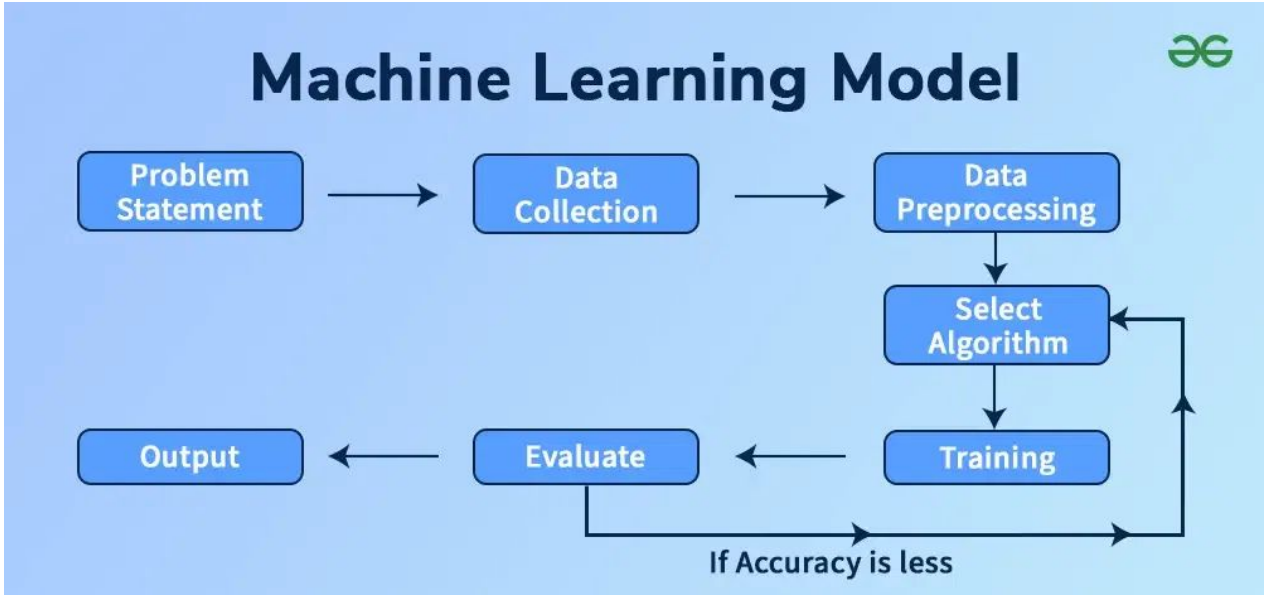
Applications and examples

- **AI algorithms:** A*, Adversarial Search, Hidden Markov Models
- **ML algorithms:** Linear Regression, Logistic Regression, K-means
- **DL algorithms:** Deep neural network, Convolutional neural network

Some cool stuff to explore

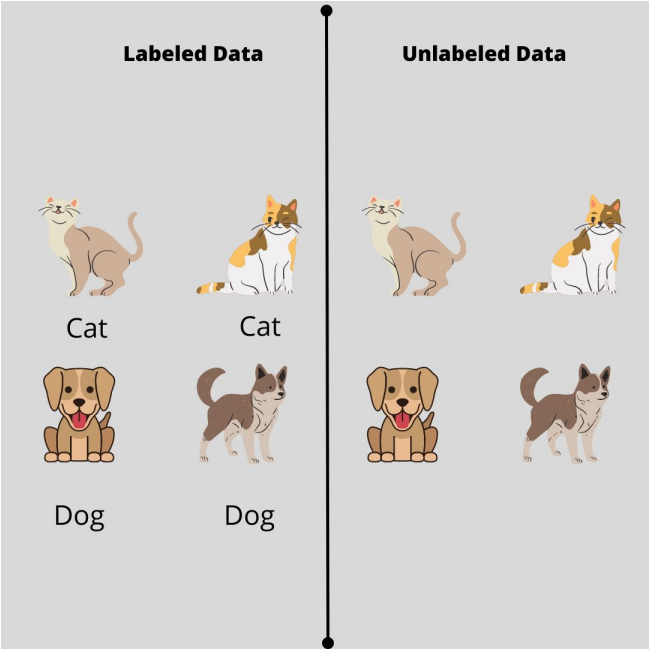
- **Segment Anything Model 3:** https://aidemos.meta.com/segment-anything/gallery
- **Waymo:** https://www.youtube.com/watch?v=B8R148hFxPw

How ML models work



Source: https://www.geeksforgeeks.org/machine-learning/machine-learning-models/

Labelled Vs Unlabelled Data



Source: https://medium.com/@varun.sja/labeled-and-unlabeled-data-what-is-the-difference-4e95e8013138

Supervised Vs Unsupervised

Step 3: Types of Machine Learning

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Supervised Learning

Most Common Approach - Model learns from labelled data where both input and correct output are provided. Like learning with a teacher who provides answers.

Unsupervised Learning

Finding Hidden Patterns - Model learns patterns from unlabelled data, discovering structure without being told what to find. Like exploring data without guidance.

Semi-Supervised Learning

Best of Both Worlds - Uses a small amount of labelled data combined with large amounts of unlabelled data, reducing labelling costs whilst maintaining accuracy.

Reinforcement Learning

Learning Through Experience - Learning by trial and error using rewards and penalties, similar to how animals learn through consequences of their actions.

Real-World Case Studies

Supervised: Fraud Detection

Credit card companies use labelled transaction data (normal vs fraudulent) to train models that identify suspicious patterns in real-time, protecting millions of customers.

Unsupervised: Customer Segmentation

Retailers analyse shopping behaviour to naturally group customers into segments like budget buyers, premium buyers, and occasional shoppers without predefined categories.

Real-World Case Studies

Semi-Supervised: Face Grouping

Google Photos requires users to label only a few faces, then automatically clusters the rest, making photo organisation effortless for billions of images.

Reinforcement: AlphaGo

Google DeepMind's AlphaGo learned Go game strategies through millions of self-play games with rewards, eventually defeating world champions.

Practice Questions

Question 1

Q: Which type of learning uses rewards and penalties?

Answer

Reinforcement Learning: the agent learns optimal behaviour through trial and error, receiving feedback in the form of rewards or penalties.

Question 2

Q: Give one example each for supervised, unsupervised, and semi-supervised learning.

Answer

Supervised: Email spam detection with labelled examples.
Unsupervised: Customer segmentation discovering natural groups.
Semi-supervised: Labelling a few product images and clustering the rest automatically.

Question 3 -

Q: Why is semi-supervised learning helpful?

Answer -

It uses only a small labelled dataset to guide learning whilst leveraging large amounts of unlabelled data, significantly reducing cost and human effort whilst still improving accuracy and generalization capabilities.

Step 4: Core Machine Learning Tasks

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Classification

Predicting Categories - Assigning inputs to discrete categories or labels. The output is always one of a fixed set of classes.

- Spam or Not Spam
- Disease or No Disease
- Cat vs Dog in images

Real Case: Medical diagnosis systems predicting COVID test results from patient symptoms and test data.

Step 4: Core Machine Learning Tasks

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Regression

Predicting Continuous Values - Predicting numerical values that can fall anywhere on a continuous scale. The output is a number, not a category.

- House price prediction
- Temperature forecasting
- Car mileage estimation

Real Case: Uber's surge pricing system predicting demand-based fare multipliers in real-time.

Step 4: Core Machine Learning Tasks

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Clustering

Grouping Similar Data - Automatically discovering natural groupings in data without predefined categories. An unsupervised task.

- Grouping customers by behaviour
- Organizing documents by topic
- Segmenting images by content

Real Case: Netflix clusters users with similar viewing preferences to power personalized recommendations.

Classification vs Regression: Key Differences

Classification -

Characteristics

- Output is a discrete category or class
- Finite number of possible outcomes
- Decision boundaries separate classes
- Examples: Yes/No, Cat/Dog/Bird, A/B/C/D grades

When to Use: When your target is a label, category, or class membership

Regression -

Characteristics

- Output is a continuous numerical value
- Infinite possible outcomes within a range
- Predicts quantities and measurements
- Examples: £250,000, 23.5°C, 45.2 mph

When to Use: When your target is a number that can vary continuously

Practice Questions

Question 1

Q: Is predicting house price a classification or regression task?

Answer

Regression. House prices are continuous numerical values that can take any value within a range, not discrete categories.

Question 2

Q: A company wants to group similar customers but has no labels. Which ML task should they use?

Answer

Clustering (Unsupervised Learning). Since there are no predefined categories or labels, clustering will automatically discover natural customer segments based on behavioural patterns.

Question 3

Q: Explain why classification cannot be used to predict stock market prices.

Answer

Stock prices are continuous numeric values that vary infinitely and smoothly over time.

Classification predicts discrete, finite labels or categories.

To predict exact price values like $142.37, you need regression, which can output any continuous number rather than just predefined classes.

Revisit...

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Congratulations on Completing the Foundation!

You've now covered the essential building blocks of machine learning:

- Understanding ML fundamentals
- Grasping the AI → ML → DL hierarchy and relationships
- Learning the four main types of learning paradigms
- Mastering the three core ML tasks and their applications

Next Steps

Keep learning, stay curious, and remember: **Machine learning is not magic** it's mathematics and patterns learned from data.

Thank You !!

See You Guys in Next Session :)